Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Shelhamer, Mark Sc.D.		
Project Title:	Context-specificity and other approaches to neurovestibular adaptation		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurovestibular Adaptation Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21205-1832	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	10/01/2000	**End Date:**	02/29/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zee, David (The Johns Hopkins University) Minor, Lloyd (The Johns Hopkins University) Angelaki, Dora (Washington University) Zhou, Wu (University of Mississippi Medical Center)		
Grant/Contract No.:	NCC 9-58-NA.002.05		
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Task Description:	We propose context-specific adaptation (CSA) as a countermeasure to some of the deleterious sensorimotor effects of space flight. By CSA we mean the ability of an organism to 1) maintain two different adapted states for a response (such as two different saccade or VOR gains), 2) have each state associated with a specific context (such as g level), and 3) switch between the adapted states immediately upon a change in context (i.e., without de-adaptation and re-adaptation upon each transition). CSA is the main theme of our project. This phenomenon can be useful during phases of space flight that require transitions between different g environments (e.g., in and out of artificial gravity, from orbital flight to planetary landing). A related theme is the determination of effective adaptation procedures and effective context cues. The project consists of an integrated set of experiments that have as their overall goal the design of a spaceflight countermeasure based on forms of vestibular adaptation. The experiments include investigations at Johns Hopkins on the effects of varying g levels on saccades and ocular alignment, the relationship between the translational VOR and smooth pursuit and the role of the cerebellum on adaptation of otolith-ocular reflexes, and adaptation of translational VORs in response to transient motions. Other experiments at Washington University (St. Louis) study how CSA might transfer between eye movements and limb movements, and experiments at the University of Mississippi Medical Center investigate adaptation of the LVOR with transient accelerations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research can be of potential benefit in the area of physical therapy and rehabilitation following loss of vestibular function. Since many such rehabilitation exercises take place in the clinic, their beneficial effects may be strongly associated with the specific environment or stimulus conditions that are present in that setting, and may not transfer readily to normal everyday settings. Recognition and better understanding of this type of context-specific behavior can aid in the design of better programs of rehabilitation.
Task Progress:	September 2003 Update: 1. Two negative findings are of interest. First, we found that there is not a noticeable vertical error in horizontal saccades in parabolic flight. We thought that there might be, based on findings of disconjugate torsion in altered g levels, and saccade errors with the eyes deviated torsionally. The fact that we did not find such an error in the altered g levels of parabolic flight suggests that we can rule out this aspect of saccade accuracy as a confounding sensorimotor issue in space flight. The other negative finding is that asymmetries in vertical saccades (gain 5 and latency) do not appear to be solely, or even predominantly, gravity-related. Again, this suggests that vertical saccades are likely not adversely affected during flight, at least as a direct consequence of altered g level. 2. A very significant and unexpected finding is the presence of vertical ocular misalignment (skew) during the altered g levels of parabolic flight. This was found in the course of the investigations on saccade error described above. It was first noticed when subjects reported that they saw a single small target light split into two (diplopia), and that they could not fuse the two images, especially in the 1.8 g phase of flight. This phenomenon may be another consequence (as disconjugate torsion likely is) of otolith asymmetry, and has clear implications for piloting and other tasks during g transitions. 3. Our various findings on the properties and adaptability of the linear VOR (LVOR) have increased our base of knowledge of this fundamental response. We hope to use this paradigm as a test of otolith and cerebellar function in the future. In particular, we feel that it can be an important part of a standard pre-flight/post-flight vestibular test battery for flight crews. Additional Note: Peer-reviewed papers published between 2003-2004 have been added to the Cumulative Bibliography for this project. (Ed., 1/11/24)
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Shelhamer M, Zee DS. "Context-specific adaptation and its significance for neurovestibular problems of space flight." J Vestibular Res. 2003 Dec 28;13(4-6):345-62. http://dx.doi.org/10.3233/VES-2003-134-618 ; PMID: 15096677 , Dec-2003
Articles in Peer-reviewed Journals	Aboukhalil A, Shelhamer M, Clendaniel R. "Acquisition of context-specific adaptation is enhanced with rest intervals between changes in context state, suggesting a new form of motor consolidation." Neuroscience letters. 2004 Oct 14;369(2):162-7. https://doi.org/10.1016/j.neulet.2004.07.085 ; PMID: 15450688 , Oct-2004
Articles in Peer-reviewed Journals	Lewis RF, Clendaniel RA, Zee DS. "Vergence-dependent adaptation of the vestibulo-ocular reflex." Exp Brain Res. 2003 Oct;152(3):335-40. https://doi.org/10.1007/s00221-003-1563-9 ; PMID: 12879175 , Oct-2003
Articles in Peer-reviewed Journals	Ramat S, Zee DS. "Ocular motor responses to abrupt interaural head translation in normal humans." J Neurophysiol. 2003 Aug;90(2):887-902. https://doi.org/10.1152/jn.01121.2002 ; PMID: 12672783 , Aug-2003
Articles in Peer-reviewed Journals	Trillenberg P, Shelhamer M, Roberts D, Zee D. "Cross-axis adaptation of torsional components in the yaw-axis vestibulo-ocular reflex." Exp Brain Res. 2003 Jan;148:158-65. https://doi.org/10.1007/s00221-002-1285-4 ; PMID: 12520403 , Jan-2003
Articles in Peer-reviewed Journals	Vogelstein JT, Snyder LH, Angelaki DE. "Accuracy of saccades to remembered targets as a function of body orientation in space." J Neurophysiol. 2003 Jul 1;90(1):521-4. https://doi.org/10.1152/jn.00141.2003 ; PMID: 12843314 , Jul-2003
Articles in Peer-reviewed Journals	Walker MF, Shelhamer M, Zee DS. "Eye-position dependence of torsional velocity during interaural translation, horizontal pursuit, and yaw-axis rotation in humans." Vision Res. 2004 Mar 1;44(6):613-20. https://doi.org/10.1016/j.visres.2003.10.009 ; PMID: 14693188 , Mar-2004

Articles in Peer-reviewed Journals	Zhou W, Weldon P, Tang B, King WM. "Rapid motor learning in the translational vestibulo-ocular reflex." J. Neurosci. 2003 May 15;23(10):4288-98. https://doi.org/10.1523/JNEUROSCI.23-10-04288.2003 ; PMID: 12764117; PMCID: PMC6741094 , May-2003